WorksheetDownload.com

10 CONCEPTUAL Q&A CLOUD · PRACTICE WORKSHEET

CONCEPTUAL SERIES

Azure Functions

Name: ______________________ Cohort/Batch: ______________________ Date: ______________________ Score: ______

AZURE FUNCTIONS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

Q1 What is Azure Functions and what is its primary purpose in its cloud ecosystem? *Threat Model*

Q2 How does IAM work with Azure Functions? *Architecture*

Q3 How do you monitor Azure Functions in production? *Configuration*

Q4 How do you scale Azure Functions to handle variable load? *Hardening*

Q5 How do you secure Azure Functions in production? *SSO / IdP*

Q6 How do you manage configuration and secrets for Azure Functions? *Observability*

Q7 What are the main cost drivers for Azure Functions and how do you optimize them? *Lifecycle*

Q8 How do you implement high availability with Azure Functions? *Compliance*

Q9 How does Azure Functions integrate with a CI/CD pipeline? *Testing*

Q10 What are common pitfalls when first adopting Azure Functions? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Azure Functions is a managed cloud service

Mitigation

Azure Functions is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A2 Access to Azure Functions is controlled via

Primitives

Access to Azure Functions is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A3 Azure Functions emits metrics to the cloud

Hardening

Azure Functions emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A4 Azure Functions supports auto-scaling policies based on

Gotchas

Azure Functions supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A5 Place Azure Functions in a private subnet

Federation

Place Azure Functions in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A6 Store sensitive values in a managed secret

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A7 Cost in Azure Functions typically comes from

Automation

Cost in Azure Functions typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A8 Deploy Azure Functions across multiple availability zones.

Governance

Deploy Azure Functions across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A9 CI/CD pipelines use the cloud CLI or

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to Azure Functions after tests pass. Infrastructure-as-code (Terraform, CDK) defines Azure Functions configuration as version-controlled code deployed via the pipeline.

A10 Common mistakes include misconfigured IAM policies (too

Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.